

TABLE F.1 Formulas for Unit Conversions*

Name, Symbol, Dimensions			Conversion Formula
Length	L	L	$1 \text{ m} = 3.281 \text{ ft} = 1.094 \text{ yd} = 39.37 \text{ in} = \text{km}/1000 = 10^6 \mu\text{m}$ $1 \text{ ft} = 0.3048 \text{ m} = 12 \text{ in} = \text{mile}/5280 = \text{km}/3281$ $1 \text{ mm} = \text{m}/1000 = \text{in}/25.4 = 39.37 \text{ mil} = 1000 \mu\text{m} = 10^7 \text{ \AA}$
Speed	V	L/T	$1 \text{ m/s} = 3.600 \text{ km/hr} = 3.281 \text{ ft/s} = 2.237 \text{ mph} = 1.944 \text{ knots}$ $1 \text{ ft/s} = 0.3048 \text{ m/s} = 0.6818 \text{ mph} = 1.097 \text{ km/hr} = 0.5925 \text{ knots}$
Mass	m	M	$1 \text{ kg} = 2.205 \text{ lbm} = 1000 \text{ g} = \text{slug}/14.59 = (\text{metric ton or tonne or Mg})/1000$ $1 \text{ lbm} = \text{lbf} \cdot \text{s}^2/(32.17 \text{ ft}) = \text{kg}/2.205 = \text{slug}/32.17 = 453.6 \text{ g}$ $= 16 \text{ oz} = 7000 \text{ grains} = \text{short ton}/2000 = \text{metric ton (tonne)}/2205$
Density	ρ	M/L^3	$1000 \text{ kg/m}^3 = 62.43 \text{ lbm/ft}^3 = 1.940 \text{ slug/ft}^3 = 8.345 \text{ lbm/gal (US)}$
Force	F	ML/T^2	$1 \text{ lbf} = 4.448 \text{ N} = 32.17 \text{ lbm} \cdot \text{ft/s}^2$ $1 \text{ N} = \text{kg} \cdot \text{m/s}^2 = 0.2248 \text{ lbf} = 10^5 \text{ dyne}$
Pressure, Shear Stress	p, τ	ML/T^2	$1 \text{ Pa} = \text{N/m}^2 = \text{kg/m} \cdot \text{s}^2 = 10^{-5} \text{ bar} = 1.450 \times 10^{-4} \text{ lbf/in}^2 = \text{inch H}_2\text{O}/2.9$ $= 0.007501 \text{ torr} = 10.00 \text{ dyne/cm}^2$ $1 \text{ atm} = 101.3 \text{ kPa} = 2116 \text{ psf} = 1.013 \text{ bar} = 14.70 \text{ lbf/in}^2 = 33.90 \text{ ft of water}$ $= 29.92 \text{ in of mercury} = 10.33 \text{ m of water} = 760 \text{ mm of mercury} = 760 \text{ torr}$ $1 \text{ psi} = \text{atm}/14.70 = 6.895 \text{ kPa} = 27.68 \text{ in H}_2\text{O} = 51.71 \text{ torr}$
Volume	V	L^3	$1 \text{ m}^3 = 35.31 \text{ ft}^3 = 1000 \text{ L} = 264.2 \text{ U.S. gal}$ $1 \text{ ft}^3 = 0.02832 \text{ m}^3 = 28.32 \text{ L} = 7.481 \text{ U.S. gal} = \text{acre-ft}/43,560$ $1 \text{ U.S. gal} = 231 \text{ in}^3 = \text{barrel (petroleum)}/42 = 4 \text{ U.S. quarts} = 8 \text{ U.S. pints}$ $= 3.785 \text{ L} = 0.003785 \text{ m}^3$
Volume Flow Rate (Discharge)	Q	L^3/T	$1 \text{ m}^3/\text{s} = 35.31 \text{ ft}^3/\text{s} = 2119 \text{ cfm} = 264.2 \text{ gal (US)}/\text{s} = 15850 \text{ gal (US)}/\text{min}$ $1 \text{ cfs} = 1 \text{ ft}^3/\text{s} = 28.32 \text{ L}/\text{s} = 7.481 \text{ gal (US)}/\text{s} = 448.8 \text{ gal (US)}/\text{min}$
Mass Flow Rate	$\dot{m}$	M/T	$1 \text{ kg/s} = 2.205 \text{ lbm/s} = 0.06852 \text{ slug/s}$
Energy and Work	E, W	ML^2/T^2	$1 \text{ J} = \text{kg} \cdot \text{m}^2/\text{s}^2 = \text{N} \cdot \text{m} = \text{W} \cdot \text{s} = \text{volt} \cdot \text{coulomb} = 0.7376 \text{ ft} \cdot \text{lbf}$ $= 9.478 \times 10^{-4} \text{ Btu} = 0.2388 \text{ cal} = 0.0002388 \text{ Cal} = 10^7 \text{ erg} = \text{kWh}/3.600 \times 10^6$
Power	$P, \dot{E}, \dot{W}$	ML^2/T^3	$1 \text{ W} = \text{J/s} = \text{N} \cdot \text{m/s} = \text{kg} \cdot \text{m}^2/\text{s}^3 = 1.341 \times 10^{-3} \text{ hp}$ $= 0.7376 \text{ ft} \cdot \text{lbf/s} = 1.0 \text{ volt-ampere} = 0.2388 \text{ cal/s} = 9.478 \times 10^{-4} \text{ Btu/s}$ $1 \text{ hp} = 0.7457 \text{ kW} = 550 \text{ ft} \cdot \text{lbf/s} = 33,000 \text{ ft} \cdot \text{lbf/min} = 2544 \text{ Btu/h}$
Angular Speed	ω	T^{-1}	$1.0 \text{ rad/s} = 9.549 \text{ rpm} = 0.1591 \text{ rev/s}$
Viscosity	μ	M/LT	$1 \text{ Pa} \cdot \text{s} = \text{kg/m} \cdot \text{s} = \text{N} \cdot \text{s}/\text{m}^2 = 10 \text{ poise} = 0.02089 \text{ lbf} \cdot \text{s}/\text{ft}^2 = 0.6720 \text{ lbm}/\text{ft} \cdot \text{s}$
Kinematic Viscosity	ν	L^2/T	$1 \text{ m}^2/\text{s} = 10.76 \text{ ft}^2/\text{s} = 10^6 \text{ cSt}$
Temperature	T	Θ	$\text{K} = ^\circ\text{C} + 273.15 = ^\circ\text{R}/1.8$ $^\circ\text{C} = (^\circ\text{F} - 32)/1.8$ $^\circ\text{R} = ^\circ\text{F} + 459.67 = 1.8 \text{ K}$ $^\circ\text{F} = 1.8^\circ\text{C} + 32$

*A useful online reference is www.onlineconversion.com

TABLE F.2 Commonly Used Equations

Ideal gas law	
$p = \rho RT$	(Eq. 1.10, p. 14)
Specific weight	
$\gamma = \rho g$	(Eq. 2.3, p. 31)
Specific gravity	
$S = \frac{\rho}{\rho_{H_2O \text{ at } 4^\circ C}} = \frac{\gamma}{\gamma_{H_2O \text{ at } 4^\circ C}}$	(Eq. 2.5, p. 32)
Kinematic viscosity	
$\nu = \mu/\rho$	(Eq. 2.15, p. 38)
Definition of viscosity	
$\tau = \mu \frac{dV}{dy}$	(Eq. 2.16, p. 39)
Pressure equations	
$p_{\text{gage}} = p_{\text{abs}} - p_{\text{atm}}$	(Eq. 3.3a, p. 62)
$p_{\text{vacuum}} = p_{\text{atm}} - p_{\text{abs}}$	(Eq. 3.3b, p. 62)
Hydrostatic equation	
$\frac{p_1}{\gamma} + z_1 = \frac{p_2}{\gamma} + z_2 = \text{constant}$	(Eq. 3.10a, p. 66)
$p_z = p_1 + \gamma z_1 = p_2 + \gamma z_2 = \text{constant}$	(Eq. 3.10b, p. 66)
$\Delta p = -\gamma \Delta z$	(Eq. 3.10c, p. 66)
Manometer equations	
$p_2 = p_1 + \sum_{\text{down}} \gamma_i h_i - \sum_{\text{up}} \gamma_i h_i$	(Eq. 3.21, p. 74)
$h_1 - h_2 = \Delta h (\gamma_B / \gamma_A - 1)$	(Eq. 3.22, p. 75)
Hydrostatic force equations (flat panels)	
$F_p = \bar{p} A$	(Eq. 3.28, p. 80)
$y_{cp} - \bar{y} = \frac{\bar{I}}{\bar{y} A}$	(Eq. 3.33, p. 81)
Buoyant force (Archimedes equation)	
$F_B = \gamma V_D$	(Eq. 3.41a, p. 87)
The Bernoulli equation	
$\left(\frac{p_1}{\gamma} + \frac{V_1^2}{2g} + z_1 \right) = \left(\frac{p_2}{\gamma} + \frac{V_2^2}{2g} + z_2 \right)$	(Eq. 4.21b, p. 133)
$\left(p_1 + \frac{\rho V_1^2}{2} + \rho g z_1 \right) = \left(p_2 + \frac{\rho V_2^2}{2} + \rho g z_2 \right)$	(Eq. 4.21a, p. 133)
Coefficient of pressure	
$C_p = \frac{p_z - p_{zo}}{\rho V_o^2 / 2} = \frac{h - h_o}{V_o^2 / (2g)}$	(Eq. 4.47, p. 147)
Volume flow rate equation	
$Q = \bar{V} A = \frac{\dot{m}}{\rho} = \int_A V dA = \int_A \mathbf{V} \cdot d\mathbf{A}$	(Eq. 5.10, p. 174)
Mass flow rate equation	
$\dot{m} = \rho A \bar{V} = \rho Q = \int_A \rho V dA = \int_A \rho \mathbf{V} \cdot d\mathbf{A}$	(Eq. 5.11, p. 174)
Continuity equation	
$\frac{d}{dt} \int_{cv} \rho dV + \int_{cs} \rho \mathbf{V} \cdot d\mathbf{A} = 0$	(Eq. 5.28, p. 183)
$\frac{d}{dt} M_{cv} + \sum_{cs} \dot{m}_o - \sum_{cs} \dot{m}_i = 0$	(Eq. 5.29, p. 183)
$\rho_2 A_2 V_2 = \rho_1 A_1 V_1$	(Eq. 5.33, p. 189)
Momentum equation	
$\sum \mathbf{F} = \frac{d}{dt} \int_{cv} \rho \mathbf{p} dV + \int_{cs} \rho \mathbf{p} \mathbf{V} \cdot d\mathbf{A}$	(Eq. 6.7, p. 213)
$\sum \mathbf{F} = \frac{d(m_{cv} \mathbf{v}_{cv})}{dt} + \sum_{cs} \dot{m}_o \mathbf{v}_o - \sum_{cs} \dot{m}_i \mathbf{v}_i$	(Eq. 6.10, p. 213)
Energy equation	
$\left(\frac{p_1}{\gamma} + \alpha_1 \frac{\bar{V}_1^2}{2g} + z_1 \right) + h_p = \left(\frac{p_2}{\gamma} + \alpha_2 \frac{\bar{V}_2^2}{2g} + z_2 \right) + h_t + h_L$	(Eq. 7.29, p. 262)
The power equation	
$P = FV = T\omega$	(Eq. 7.3, p. 255)
$P = \dot{m}gh = \gamma Qh$	(Eq. 7.31, p. 265)
Efficiency of a machine	
$\eta = \frac{P_{\text{output}}}{P_{\text{input}}}$	(Eq. 7.32, p. 267)
Reynolds number (pipe)	
$Re_D = \frac{VD}{\nu} = \frac{\rho VD}{\mu} = \frac{4Q}{\pi D \nu} = \frac{4\dot{m}}{\pi D \mu}$	(Eq. 10.1, p. 361)
Combined head loss equation	
$h_L = \sum_{\text{pipes}} f \frac{L}{D} \frac{V^2}{2g} + \sum_{\text{components}} K \frac{V^2}{2g}$	(Eq. 10.45, p. 382)
Friction factor f (Resistance coefficient)	
$f = \frac{64}{Re_D} \quad Re \leq 2000$	(Eq. 10.34, p. 370)
$f = \frac{0.25}{\left[\log_{10} \left(\frac{k_s}{3.7D} + \frac{5.74}{Re_D^{0.9}} \right) \right]^2} \quad (Re \geq 3000)$	(Eq. 10.39, p. 375)
Drag force equation	
$F_D = C_D A \left(\frac{\rho V_o^2}{2} \right)$	(Eq. 11.5, p. 409)
Lift force equation	
$F_L = C_L A \left(\frac{\rho V_o^2}{2} \right)$	(Eq. 11.17, p. 424)

TABLE F.3 Useful Constants

Name of Constant	Value
Acceleration of gravity	$g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$
Universal gas constant	$R_u = 8.314 \text{ kJ/kmol} \cdot \text{K} = 1545 \text{ ft} \cdot \text{lb}/\text{lbmol} \cdot ^\circ\text{R}$
Standard atmospheric pressure	$p_{\text{atm}} = 1.0 \text{ atm} = 101.3 \text{ kPa} = 14.70 \text{ psi} = 2116 \text{ psf} = 33.90 \text{ ft of water}$ $p_{\text{atm}} = 10.33 \text{ m of water} = 760 \text{ mm of Hg} = 29.92 \text{ in of Hg} = 760 \text{ torr} = 1.013 \text{ bar}$

TABLE F.4 Properties of Air [$T = 20^\circ\text{C}$ (68°F), $p = 1 \text{ atm}$]

Property	SI Units	Traditional Units
Specific gas constant	$R_{\text{air}} = 287.0 \text{ J/kg} \cdot \text{K}$	$R_{\text{air}} = 1716 \text{ ft} \cdot \text{lb}/\text{slug} \cdot ^\circ\text{R}$
Density	$\rho = 1.20 \text{ kg/m}^3$	$\rho = 0.0752 \text{ lbm/ft}^3 = 0.00234 \text{ slug/ft}^3$
Specific weight	$\gamma = 11.8 \text{ N/m}^3$	$\gamma = 0.0752 \text{ lbf/ft}^3$
Viscosity	$\mu = 1.81 \times 10^{-5} \text{ N} \cdot \text{s/m}^2$	$\mu = 3.81 \times 10^{-7} \text{ lbf} \cdot \text{s/ft}^2$
Kinematic viscosity	$\nu = 1.51 \times 10^{-5} \text{ m}^2/\text{s}$	$\nu = 1.63 \times 10^{-4} \text{ ft}^2/\text{s}$
Specific heat ratio	$k = c_p/c_v = 1.40$	$k = c_p/c_v = 1.40$
Specific heat	$c_p = 1004 \text{ J/kg} \cdot \text{K}$	$c_p = 0.241 \text{ Btu/lbm} \cdot ^\circ\text{R}$
Speed of sound	$c = 343 \text{ m/s}$	$c = 1130 \text{ ft/s}$

TABLE F.5 Properties of Water [$T = 15^\circ\text{C}$ (59°F), $p = 1 \text{ atm}$]

Property	SI Units	Traditional Units
Density	$\rho = 999 \text{ kg/m}^3$	$\rho = 62.4 \text{ lbm/ft}^3 = 1.94 \text{ slug/ft}^3$
Specific weight	$\gamma = 9800 \text{ N/m}^3$	$\gamma = 62.4 \text{ lbf/ft}^3$
Viscosity	$\mu = 1.14 \times 10^{-3} \text{ N} \cdot \text{s/m}^2$	$\mu = 2.38 \times 10^{-5} \text{ lbf} \cdot \text{s/ft}^2$
Kinematic viscosity	$\nu = 1.14 \times 10^{-6} \text{ m}^2/\text{s}$	$\nu = 1.23 \times 10^{-5} \text{ ft}^2/\text{s}$
Surface tension (water-air)	$\sigma = 0.073 \text{ N/m}$	$\sigma = 0.0050 \text{ lbf/ft}$
Bulk modulus of elasticity	$E_v = 2.14 \times 10^9 \text{ Pa}$	$E_v = 3.10 \times 10^5 \text{ psi}$

TABLE F.6 Properties of Water [$T = 4^\circ\text{C}$ (39°F), $p = 1 \text{ atm}$]

Property	SI Units	Traditional Units
Density	$\rho = 1000 \text{ kg/m}^3$	$\rho = 62.4 \text{ lbm/ft}^3 = 1.94 \text{ slug/ft}^3$
Specific weight	$\gamma = 9810 \text{ N/m}^3$	$\gamma = 62.4 \text{ lbf/ft}^3$

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Table 1.3
Conversion Factors from BG and EE Units to SI Units^a

	To Convert from	to	Multiply by
Acceleration	ft/s ²	m/s ²	3.048 E - 1
Area	ft ²	m ²	9.290 E - 2
Density	lbm/ft ³	kg/m ³	1.602 E + 1
	slugs/ft ³	kg/m ³	5.154 E + 2
Energy	Btu	J	1.055 E + 3
	ft · lb	J	1.356
Force	lb	N	4.448
Length	ft	m	3.048 E - 1
	in.	m	2.540 E - 2
	mile	m	1.609 E + 3
Mass	lbm	kg	4.536 E - 1
	slug	kg	1.459 E + 1
Power	ft · lb/s	W	1.356
	hp	W	7.457 E + 2
Pressure	in. Hg (60 °F)	N/m ²	3.377 E + 3
	lb/ft ² (psf)	N/m ²	4.788 E + 1
	lb/in. ² (psi)	N/m ²	6.895 E + 3
Specific weight	lb/ft ³	N/m ³	1.571 E + 2
Temperature	°F	°C	$T_C = (5/9)(T_F - 32°)$
	°R	K	5.556 E - 1
Velocity	ft/s	m/s	3.048 E - 1
	mi/hr (mph)	m/s	4.470 E - 1
Viscosity (dynamic)	lb · s/ft ²	N · s/m ²	4.788 E + 1
Viscosity (kinematic)	ft ² /s	m ² /s	9.290 E - 2
Volume flowrate	ft ³ /s	m ³ /s	2.832 E - 2
	gal/min (gpm)	m ³ /s	6.309 E - 5

^aFor more unit conversions, refer to Appendix E.

Table 1.4
Conversion Factors from SI Units to BG and EE Units^a

	To Convert from	to	Multiply by
Acceleration	m/s ²	ft/s ²	3.281
Area	m ²	ft ²	1.076 E + 1
Density	kg/m ³	lbm/ft ³	6.243 E - 2
	kg/m ³	slugs/ft ³	1.940 E - 3
Energy	J	Btu	9.478 E - 4
	J	ft · lb	7.376 E - 1
Force	N	lb	2.248 E - 1
Length	m	ft	3.281
	m	in.	3.937 E + 1
	m	mile	6.214 E - 4
Mass	kg	lbm	2.205
	kg	slug	6.852 E - 2
Power	W	ft · lb/s	7.376 E - 1
	W	hp	1.341 E - 3
Pressure	N/m ²	in. Hg (60 °F)	2.961 E - 4
	N/m ²	lb/ft ² (psf)	2.089 E - 2
	N/m ²	lb/in. ² (psi)	1.450 E - 4
Specific weight	N/m ³	lb/ft ³	6.366 E - 3
Temperature	°C	°F	$T_F = 1.8 T_C + 32°$
	K	°R	1.800
Velocity	m/s	ft/s	3.281
	m/s	mi/hr (mph)	2.237
Viscosity (dynamic)	N · s/m ²	lb · s/ft ²	2.089 E - 2
Viscosity (kinematic)	m ² /s	ft ² /s	1.076 E + 1
Volume flowrate	m ³ /s	ft ³ /s	3.531 E + 1
	m ³ /s	gal/min (gpm)	1.585 E + 4

^aFor more unit conversions, refer to Appendix E.